The Basics

The aim of this Quick Start Guide

The aim of this guide is to quickly get you creating, and running *Test Programs* on a *need-to-know* basis. I will only tell you what you really need-to-know to create, and run an *em**Power**™ Test Program*. To do this, we (that's you and me) are going to:

1. Start *em**Power**™*, and make sure it is *Offline*, and that the demonstration *Configuration* file `Demo.cfg` is the *Current Configuration* file
2. Go through all the steps to create a complete *em**Power**™ Test Program* using *em**Power**™ Assistant*, and ***Power**EDIT*, for a DC-to-DC converter (specifications on page 13). This is an *Offline* (computer only, simulated hardware interaction) exercise, and is a useful exercise for all *em**Power**™* developers no matter what kind of device you want to use *em**Power**™* to test
3. Then we'll use ***Power**TEST* to run the *Test Program*

The system, briefly

The *em**Power**™* system has three main components:

- *TEST* – Used to run, or execute *Test Programs*
- *EDIT* – Used to create *Test Programs*
- *CONTROL* – Used to interactively control: *Input*, *Output*, and *Measurement* devices

In this guide, we will only be looking at *EDIT*, and *TEST* in any detail.

Key concepts, and terms

Most concepts and terms in this guide are called out and explained when they are first mentioned, but there are a few that you need to know before going any further, they are:

- *UUT* – Unit Under Test, in this document our UUT is a DC-DC converter
- *Off-Line Testing* – The running of an *em**Power**™ Test Program* on a computer with simulated hardware interaction
- *Input* – Where the UUT gets it's power from
- *Output* – Where the UUT's output goes
- *Nominal* – The normally expected value
- *Vector* – A *Vector* is a snapshot of a device state. This snapshot contains the settings of that device for a specific point in the run of a *Test Program*. The *em**Power**™* system has: *Input Vectors*, *Output Vectors*, and *Measurement Vectors*.

Formats for values

The input fields in *em**Power***™ accept values in decimal and *Engineering Notation* formats.

Examples:

1. If *48 volts* is required, you may enter either: *48v*, *48*, *0.048kv*, or *0.48k* etc.
2. If a *100 milli-second* delay is required, you may enter either: *100m*, *100ms*, *0.1s*, or *0.1* etc.

Letter	Units	Decimal Notation	Engineering Notation
p or P	pico	0.000000000001	1E-12
n or N	nano	0.000000001	1E-9
u or U	micro	0.000001	1E-6
m (case sensitive)	milli	0.001	1E-3
k or K	kilo	1,000	1E3
M (case sensitive)	mega	1,000,000	1E6
g or G	giga	1,000,000,000	1E9

Table 1 - Formats for values

The partial *Model 4805 DC-to-DC Converter* Specifications

Typical at T_{BP} - 25°C, nominal line and 75% load, unless otherwise specified.

PARAMETER	MIN.	TYP.	MAX.	UNITS	TEST CONDITONS
Input Characteristics					
Voltage	42	48	60	Volts	
Output Characteristics					
Voltage		5		Volts	
Current			40	Amps	
Power			200	Watts	
Setpoint accuracy			2%	V_{NOM}	
Load/line regulation			0.5%	V_{NOM}	LL to HL, 10% to Full Load
Load/line regulation			1%	V_{NOM}	LL to HL, No Load to 10%
Output ripple - pp: (5V)			5%		20 MHz bandwidth
OVP set point		125%		V_{NOM}	Recycle power
Current limit	105%		135%	I_{NOM}	Automatic restart
Short circuit current	20%		140%	I_{NOM}	
Thermal Characteristics					
Efficiency		78-88%			

Table 2 - The partial *Model 4805 DC-to-DC Converter* Specifications

Starting *em**Power***™

Step 1 – Starting

The *em**Power***™ shortcut icon below, should be on your desktop:

Figure 1 – The *emPower***™ **desktop icon**

- ⇨ Double-click the icon to start *em**Power***™

Step 2 – Logging on

The first time you run *em**Power***™ there are no users. The first user should logon at the *highest User Level - Administrator*. For the purpose of this guide, you can logon at this *User Level* with no password.

- ⇨ In the *User Name* entry box, type-in: *Administrator*
- ⇨ Click the *Logon* button – you will be prompted for passwords
- ⇨ Enter the passwords, or click the *OK* button for no passwords – *em**Power***™ will start, showing the full screen image, Figure 2

Figure 2 – The *emPower***™ **screen**

Step 3 – Going to *Offline* mode

With *em**Power***™ running

- ⇨ Click the *Utilities* button in the *em**Power***™ task bar, Figure 3, – the *em**Power***™ *Utilities* dialog window (Options tab) opens

Figure 3 – The *emPower***™ task bar**

- ⇨ Click the *Hardware* tab – the *em**Power***™ *Utilities* (Hardware tab) dialog window, Figure 4, opens

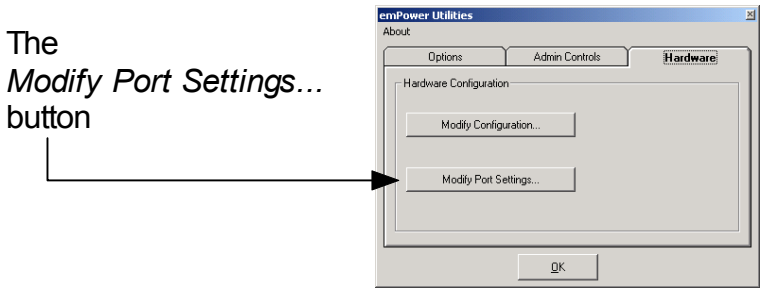

Figure 4 – The *emPower***™ *Utilities* dialog window (*Hardware* tab)**

- ⇨ Click the *Modify Port Settings*… button – the *NHR Port Configuration Utility* dialog window, Figure 5, opens

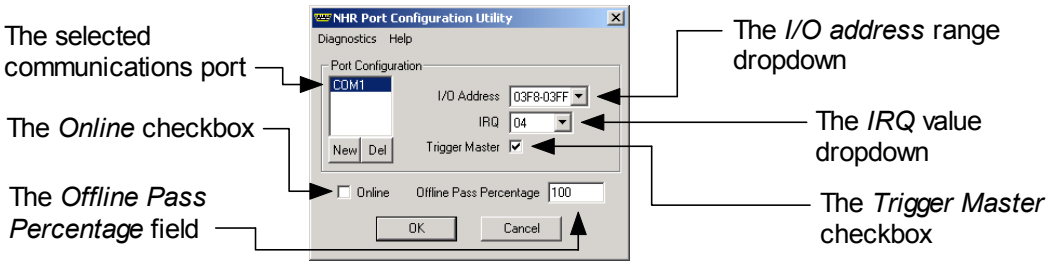

Figure 5 – The *NHR Port Configuration Utility* dialog window

There is much more to the *NHR Port Configuration Utility* than we need to go into now.

- ⇨ Make sure that the *Online* checkbox is unchecked – uncheck it if it isn't
- ⇨ Click the *OK* button – *em**Power***™ is in *Offline* mode